JS Astrophysics Computational Lab: Assignment 3

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1 Abstract

The close approaches of nearby stars (Stellar flybys) can penetrate the Oort Cloud, which is a spherical shell around the solar system of comets ranging from 0.01–0.5 pc. This can cause comets to be displaced from their orbits and hurled into the solar system due to the stars gravity. This lab uses the Gaia Catalogue of Nearby Stars (GCNS, containing 167020 objects) and the UltraCool Sheet (containing 3890 objects) to identify and analyse stellar flybys within a timeframe of ± 5 Myr from present day. Perihelion times and distances were computed using the approximations outlined in Bailer-Jones et al. 2015, and their uncertainties were calculated using the Monte Carlo method with 1000 iterations. Looking within the timescale of ± 5 Myr, 44 stars pass within Proxima Centauri's current distance from the Sun of (1.3 pc), 4 enter the outer Oort Cloud (< 0.48 pc), and none go past the inner Oort Cloud (< 0.0097 pc). The closest flyby is predicted to occur in 1.294 ± 0.022 Myr, it is Source ID: 4270814637616488064 and has will have a perihelion distance of 0.051 ± 0.003 pc, which is well within the Oort cloud. The relative impulse imparted to the Oort Cloud comets comes vastly from the closest flyby. This is because of the d_{ph}^{-2} dominance in the equation. The closest approaching ultracool dwarf is Scholz's Star (WISE J072003.20−084651.2) which got within the Oort Cloud (0.330 pc) approximately 80000 years ago.

2 Introduction

This lab analyses the stars within the solar neighbourhood using the Gaia Catalogue of Nearby Stars (GCNS) and the Ultracool Sheet. Each star moves with its own velocity and over millions of years these stars can come close to the Sun. These close approaches are called stellar flybys and can cause disturbances in other objects such as the Oort cloud. This is a huge spherical cluster of comets that surrounds the solar system in a range of about 0.01 - 0.5 pc (Oort et al. 1950). Comets from the Oort cloud could be sent into the solar system when their orbits are disturbed by these stellar flybys (Rickman et al. 2014, Davis et

al. 1984). Therefore, knowing what stars will have a close approach and how they will affect the Oort cloud is important for understanding the long term flux of comets that may affect Earth.

The trajectory of each star is approximated using linear motion formulations from Bailer-Jones et al. 2015. From these, the time of closest approach t_{ph} (perihelion time) and the distance of its closest approach d_{ph} (perihelion distance) can be calculated from a stars parallax, radial velocity, and proper motion. The relative impurse imparted by the passing star to Oort cloud comets is approximated using $M/(v_{ph} \quad d_{ph}^2)$ (Opik et al. 1932 and Oort et al. 1950). The d_{ph}^{-2} dominance within the equation means that the closest approaches are the most significant, rather then the mass of the star.

The Gaia mission had provided data from thousands of nearby stars, including their parallaxes, proper motions, and radial velocities. The GCNS holds all of this data within 100 pc of the Sun. The Gaia radial spectrometer collects data at optical wavelengths, so it misses a lot of cool, faint stars within near infrared wavelengths.

To combat this problem, the UltraCool Sheet (Best et al. 2020) was used. This contains data on over 3000 ultracool dwarfs of spectral types M, L, T and Y. This data is collected fro ground based infrared observations.

This report is split into 3 parts, parts 1 and 2 use the GCNS to identify the closest stellar encounters and calculate their impulse on the Oort Cloud. Part 3 focuses on the UltraCool Sheet, where the same analysis is done on the low mass, cool, faint dwarfs missed by Gaia. Uncertainties of all values not provided in the datasets is calculated using the Monte Carlo method. Where inputs of parallax, radial velocity, proper motion are sampled from Gaussian distributions, then outputs of the perihelion values are collected and their means and standard deviations are extracted.

3 Experimental Details

Part 1:
The Gaia Catalogue of Nearby Stars dataset was used in Parts 1 and 2. It contains information on stars within 100 parsecs of the Sun from Gaia data release 3. The file imported was '100pc_rvnotnull.fits', which was edited beforehand for ease of use. The file was read into Python with the astropy.io.fits library.
The parallax and its error, mean magnitudes in BP, RP and G bands along with the flux to noise ratio for each band were extracted from the file. The radial velocity, proper motion right ascension and declination along with their errors were also extracted. The distances to each star were computed by inverting the parallax ($d = 1000/\varpi$ pc), where ϖ is the parallax in mas. The absolute magnitude G was calculated from ($M_G = G - 5\log_{10}(d) + 5$). The BP - RP

colour was calculated from the mean BP and RP magnitudes, this was the x axis of the HR diagram (Figure 1).
The BP, RP and G uncertainties were calculated from the Gaia flux signal to noise ratios from the relation Error = $2.5/(\ln(10) \times \text{SNR})$, where SNR is the flux divided by its uncertainty which was extracted from the file.
The time and distance of the perihelion (closest approach) for each star was calculated using the formulation in Bailer-Jones et al. 2015. Therefore these equations for perihelion time and distance were used;

$$t_{ph} = -c_1 \frac{1}{\varpi} \frac{v_r}{v^2} \quad \text{(yr)}$$

$$d_{ph} = 10^3 \frac{1}{\varpi} \frac{v_T}{v} \quad \text{(pc)}$$

, where v_r is the radial velocity, $v_T = c_2 \sqrt{\mu_{\alpha*}^2 + \mu_\delta^2}/\varpi$ is the transverse velocity from proper motion in RA and Dec, and $v = \sqrt{v_T^2 + v_r^2}$ is the total velocity. Positive values of t_{ph} are perihelions in the future, and negative values are perihelions that have already happened.
The uncertainties in the measurements were propagated through their subsequent equations in the Monte Carlo method. This had 1000 iterations, each one included randomising samples from Gaussian distributions centred on the measured values (parallax, radial velocity, proper motion RA and Dec). The perihelion equations were applied to the samples using NumPy arrays. A sum and a sum of squares of t_{ph} and d_{ph} was calculated instead of storing all 1000 values to save on memory and run time. The mean and standard deviations were found afterwards. Input and output histograms for 3 sources within the catalogue were plotted to check for Gaussian distributions.
After error bars were plotted, some were so big it made the plots unreadable. To combat this, error bars over values higher then 100% of the measured value were removed. The measured values were still plotted, this made the plots readable. Versions of the plots without error bars are also included in the report.

Part 2:
Stellar masses were estimated by interpolating from the Mamajek relations table (Pecault & Mamajek et al. 2013). This is a table of stellar parameters that include effective temperature T_{eff} (Teff), BP-RP colour (Bp-Rp) and mass (Msun). This was read in as a txt file 'Mamajek_Relations.txt', '...' and '.....' entries were counted as NaN values.
Mass was estimated by interpolation of the Mamajek T_{eff} mass relation using 'scipy.interpolate.interp1d'. For stars with no T_{eff} estimate, the mass was estimated from another interpolation on the Mamajek BP-RP mass relation. An array for the masses was created where the temperature based estimates were added unless there was a NaN, then the colour based estimates were used.
The relative impulse of each stellar flyby on Oort cloud comets was calculated by the approximation developed in Opik et al. 1932 and Oort et al. 1950.

$$I \propto \frac{M}{v_{ph}\, d_{ph}^2}$$

where M is the stellar mass, v_{ph} is the mean perihelion velocity from the Monte Carlo method and d_{ph} is the mean perihelion distance. Due to these factors, a star thats more massive, moving slower and are closer to the sun will cause more impulse to the comets. The distance is the largest contributing factor as it is squared in the equation.

Part 3:
The Ultracool sheet (Best et al. 2020) was analysed in this part. It is a catalogue of ultracool dwarfs (M6 and later), which includes M, L, T and Y dwarfs and their planet companions. The majority of these arent in the GCNS because their cool temperatures (¡ 2500K) cause them to emit very little light in the optical wavelengths, which is where Gaia operates. The catalogue was downloaded from a google spreadsheet as a csv file and read into the python file using the 'pandas' library.
The required data was extracted using the 'pd.to_number' function. The data extracted was from the '_formula' set within the sheet, this included the parallax ('plx_formula'), radial velocity ('rv_formula), and proper motion in RA and Dec ('pmra_formula', 'pmdec_formula'). Values in the MKO J and K bands ('J_MKO', 'K_MKO') and the spectral type indexs ('sptnum_formula', M6 = 6, L0 = 10, T0 = 20, Y0 = 30) were also extracted. Absolute J band magnitudes were calculated from parallax and apparent magnitudes using the same methods in Part 1.
A colour magnitude diagram was plotted using the absolute J band magnitude against the $J-K$ colour. The points were colour coded according to what spectral type they were. The perihelion times and distances were then calculated using the same methods in Part 1 and the Monte Carlo method was reused to find the error. Errors above 100% were once again dropped. The closest approaching source was found from the d_{ph} estimates and it was overplotted on the colour magnitude diagram to allow for an estimate on its spectral type to be made.

4 Results and Analysis

Part 1

The GCNS file `100pc_rvnotnull.fits` has 167020 sources. Faint late type M dwarf stars whos apparent magnitudes are too low for Gaia to get a proper spectrum are missing when compared to the original GCNS catalogue that was looked at in Lab 10.
Figure 1 is the Hertzsprung-Russell diagram of this sample. It plots the absolute G band magnitude M_G against the BP - RP colour index. The main sequence is clearly visible, going from hot luminous stars in the upper left to

the cool, faint red stars in the lower right. Giant and subgiant stars are displaced from the main sequence at low M_G. There is a lacking of stars at the faint red end of the main sequence when compared to the diagram in Lab 10, this is because the Gaia radial velocity measurement system struggles to pick them up.

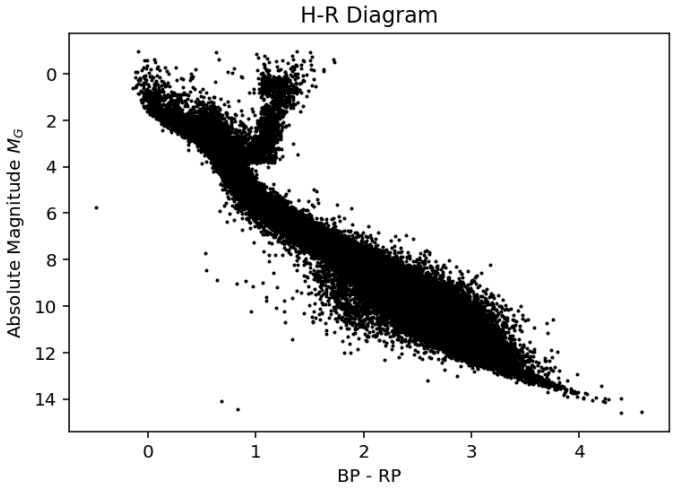


Figure 1: HR diagram of absolute G band magnitude M_G against BP - RP colour. The main sequence, giant branch, and a couple of white dwarfs in the lower left are seen.

Monte Carlo error propagation was used to determine the perihelion time t_{ph} and perihelion distance d_{ph} for every star and their uncertainties. Figures 2-4 show the input and output histograms for three sources within the catalogue (Index = 0, 300, and 1000). The 2 inputs (parallax and radial velocity) are sampled from Gaussian distributions centred on their values. The outputs of t_{ph} and d_{ph} all show Gaussian distributions.

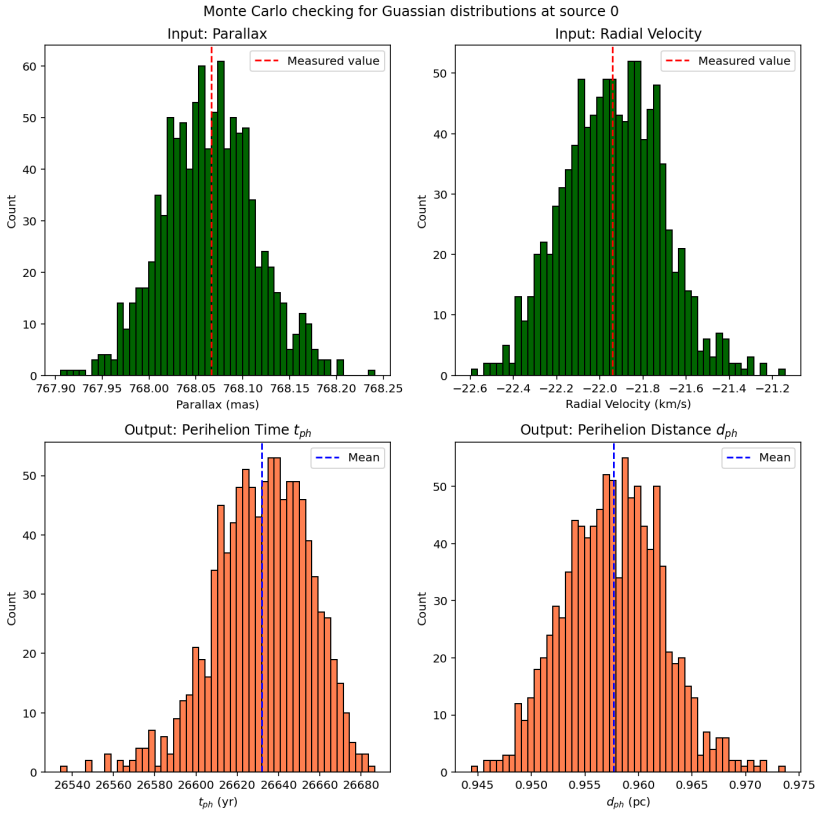


Figure 2: Monte Carlo sampling histograms for catalogue source 0. The green plots are input distributions for parallax and radial velocity. The orange plots are output distributions of the perihelion time (t_{ph}) and perihelion distance (d_{ph}). All are approximately Gaussian.

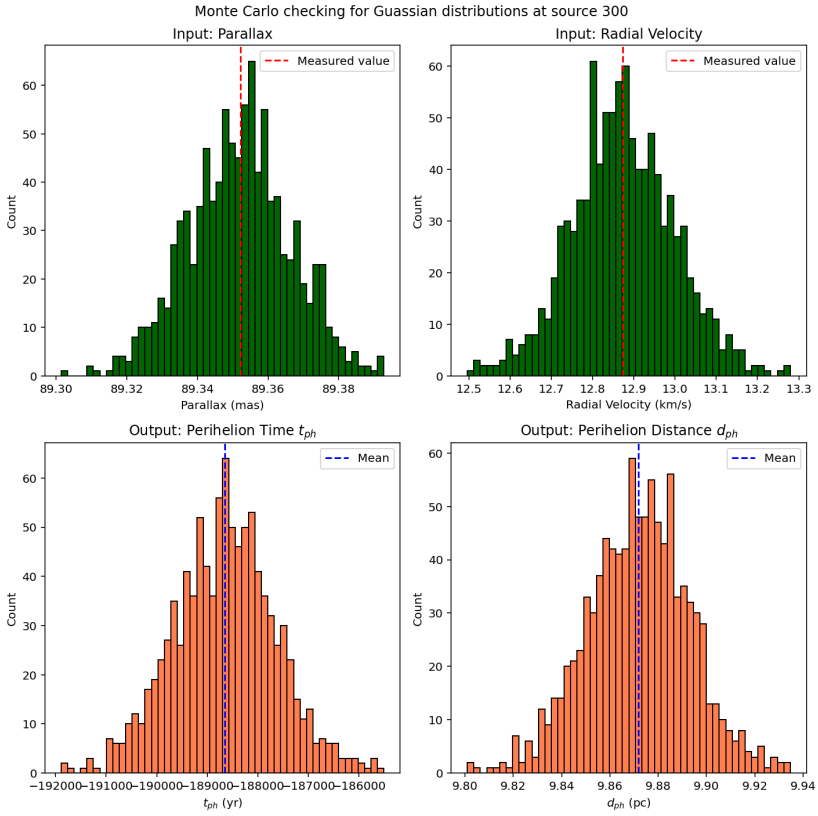


Figure 3: Monte Carlo sampling histograms for catalogue source 300. The green plots are input distributions for parallax and radial velocity. The orange plots are output distributions of the perihelion time (t_{ph}) and perihelion distance (d_{ph}). All are approximately Gaussian.

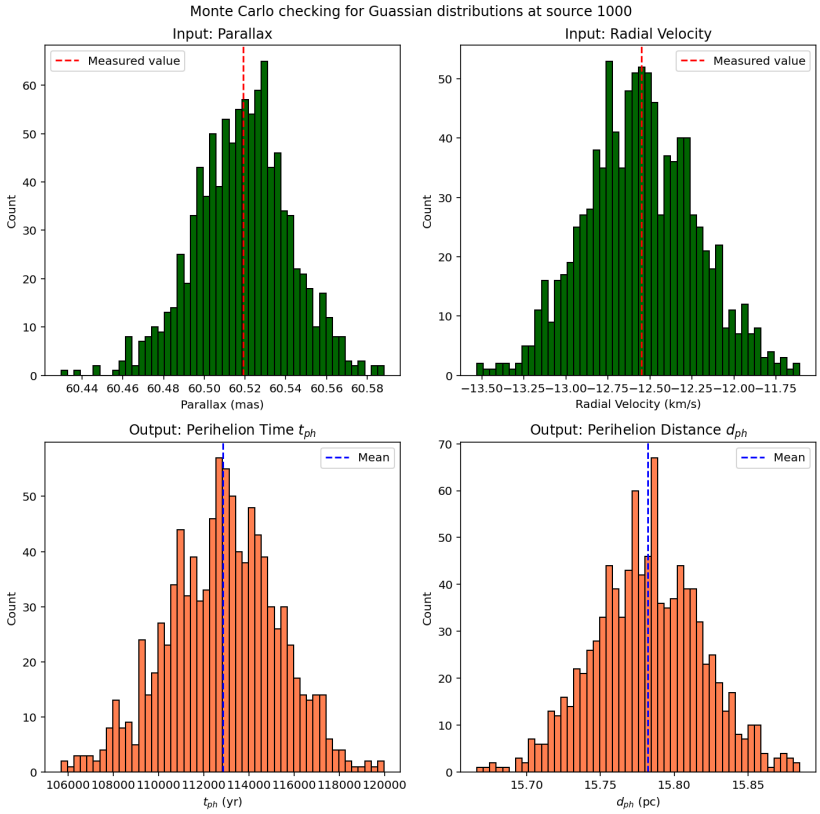


Figure 4: Monte Carlo sampling histograms for catalogue source 1000. The green plots are input distributions for parallax and radial velocity. The orange plots are output distributions of the perihelion time (t_{ph}) and perihelion distance (d_{ph}). All are approximately Gaussian.

Figures 5–6 show the distribution of perihelion times and distances for all stellar flybys within 10 pc, one plot excluding error bars and one including error bars. Figures 5–8 shows the same distributions but for stellar flybys within 2 pc, also one without error bars. Reference lines are overlayed to show the inner Oort Cloud (0.0097 pc), the outer Oort Cloud (0.48 pc) and the distance to Proxima Centauri (1.3 pc). The t_{ph} values are roughly symmetric around 0 which shows that the catalogue has an even number of past and future events. There is a number of stars within the outer Oort Cloud, showing that comets being affected by flybys are a very real event that deserves attention. Error bars are only shown for sources that have errors of less than 100% of the measured value, this is for readability purposes.

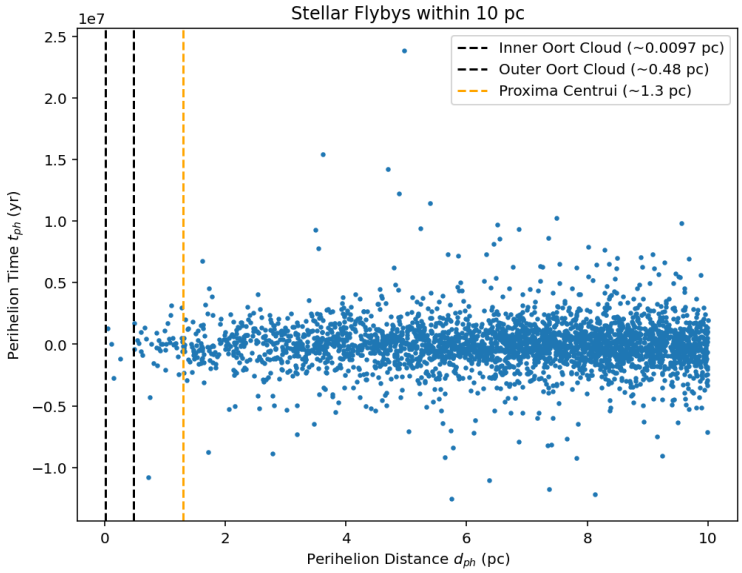


Figure 5: Perihelion time t_{ph} versus perihelion distance d_{ph} within $d_{ph} < 10\,\mathrm{pc}$, without error bars. With inner Oort Cloud, outer Oort Cloud, and the distance to Proxima Centauri.

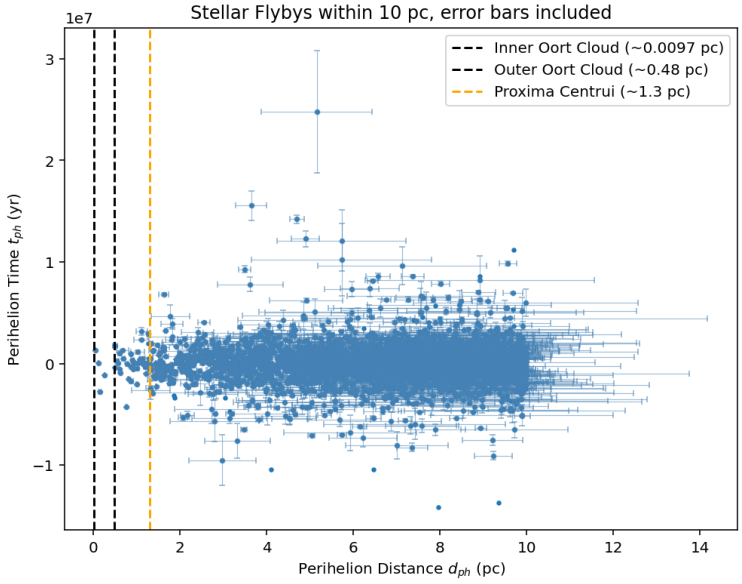


Figure 6: The same as Figure 5, but with error bars included. Stars with small d_{ph} seem to have well identified orbits, likely because of their large parallaxes

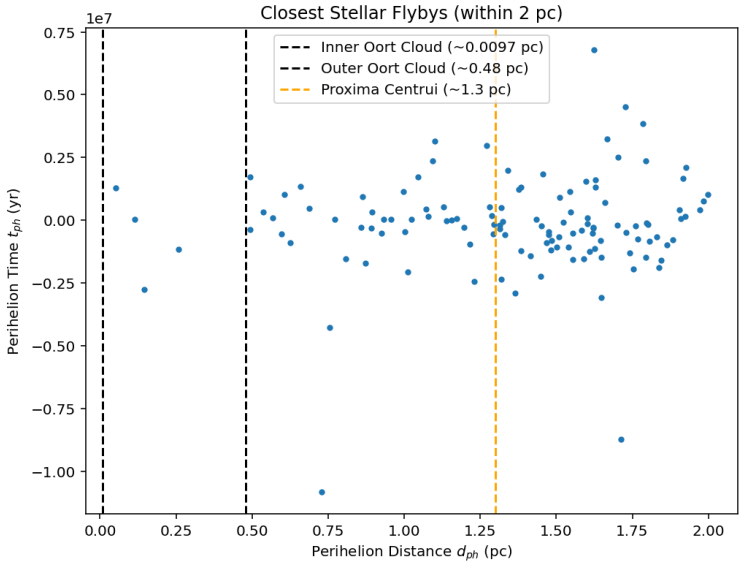


Figure 7: Zoomed view of perihelion time versus distance within $d_{ph} < 2\,\mathrm{pc}$ without error bars. Four stars are within the outer Oort Cloud none seem to be within the inner Oort Cloud.

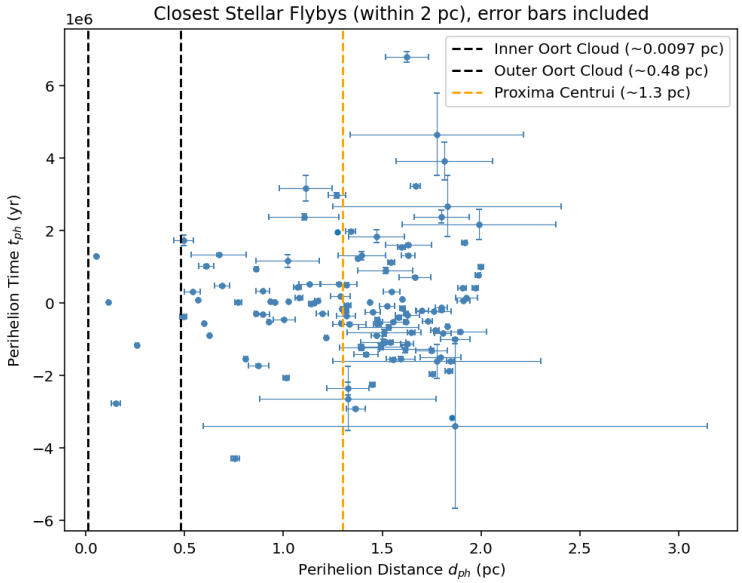


Figure 8: Same as Figure 7 but with error bars included. The error bars are generally smaller the closer you get to the origin. This shows Gaia is very precise with large parallax measurements.

Within ±5 Myr of present day 44 stars are closer than Proxima Centauri (1.3 pc) 4 stars are within outer Oort Cloud (< 0.48 pc) 0 stars are within inner Oort Cloud (< 0.0097 pc) within ±5 Myr.

Table 1 lists the ten stars with the smallest perihelion distance in the GCNS sample.

Table 1: The ten closest approaching stars by increasing d_{ph}. Current distances d are computed by inverting the parallax. Perihelion values are Monte Carlo means.

Rank	Gaia DR3 Source ID	d (pc)	t_{ph} (yr)	d_{ph} (pc)
1	4270814637616488064	19.085	$+1.295 \times 10^6 \pm 2.24e+04$	0.051±0.003
2	5544743925212648320	11.172	$+2.923 \times 10^4 \pm 6.47e+02$	0.114±0.003
3	510911618569239040	75.690	$-2.766 \times 10^6 \pm 2.32e+04$	0.151±0.024
4	5571232118090082816	97.714	$-1.158 \times 10^6 \pm 5.73e+03$	0.258±0.006
5	929788371508812288	73.005	$-3.656 \times 10^5 \pm 1.30e+04$	0.495±0.019
6	3372104035275483392	83.482	$+1.729 \times 10^6 \pm 1.41e+05$	0.496±0.048
7	4116451378388951424	84.278	$+3.122 \times 10^5 \pm 1.02e+04$	0.541±0.038
8	1952802469918554368	7.047	$+8.257 \times 10^4 \pm 1.76e+02$	0.569±0.001
9	3054509410098672000	40.060	$-5.567 \times 10^5 \pm 2.79e+03$	0.598±0.004
10	3600338081985998080	38.596	$+1.017 \times 10^6 \pm 5.31e+04$	0.608±0.040

Figure 9 shows them ten sources overplotted on the H-R diagram. The majority of them are in the main sequence between 1 and 3.5 BP - RP and from 4 - 12 M_G between roughly 7 and 12. The majority of which were around the mid to lower main sequence however, so likely K and M dwarfs. There is one outlier at BP - RP ≈ 0.8 and $M_G \approx 14$ which is well below the main sequence, this could be a white dwarf.

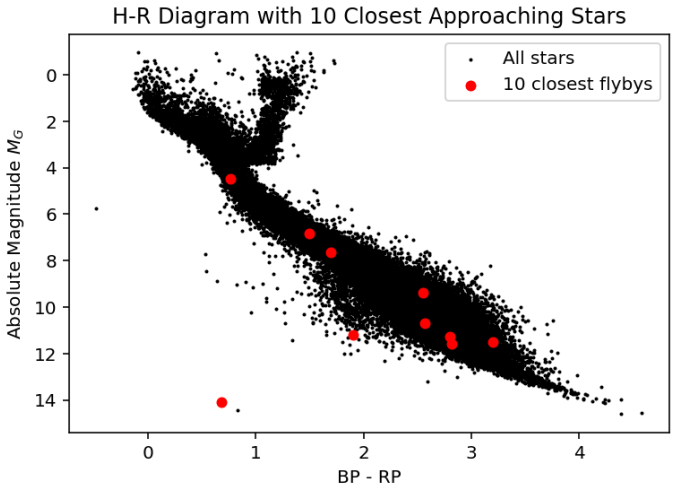


Figure 9: H-R diagram from above with the ten closest approaching stars highlighted in red. Most are K or M dwarf stars. With one possible white dwarf.

For task 8 I wanted to look at Kepler-444 as I find it very interesting since its such an ancient star with many planets. Its a K0 dwarf roughly 36.5 pc from the Sun with hosting five planets smaller than Earth. It is 11.2 billion years old which makes it one of the oldest known planetary systems. Its in Gaia DR3 so wasnt able to be found in this lab, therefore its parametric data was found manually and hardcoded. This was then used to calculate the perihelion time ($t_{ph} = +1.60 \times 10^5$ yr) and distance ($d_{ph} = 24.38$ pc). This means its closest approach will occur in 160,000 years at 24.4 pc from the Sun. This is well outside the Oort Cloud. This means Kepler-444 will have almost no impulse on Oort Cloud comets.

Part 2

Figure 10 shows the stellar mass against effective temperature $T_{\rm eff}$ for the Mamajek sequence (Pecaut & Mamajek 2013). The curve is smooth and constantly increasing from low mass dwarfs ($T_{\rm eff} \approx 2000\,{\rm K}$, $M \approx 0.1\,M_\odot$) to massive stars ($T_{\rm eff} > 35000\,{\rm K}$, $M > 25\,M_\odot$). The low mass side is very densely populated because the solar neighbourhood has mostly low mass stars. A linear interpolation of this relation was done with 'scipy.interpolate.interp1d', which gave masses to all GCNS stars.

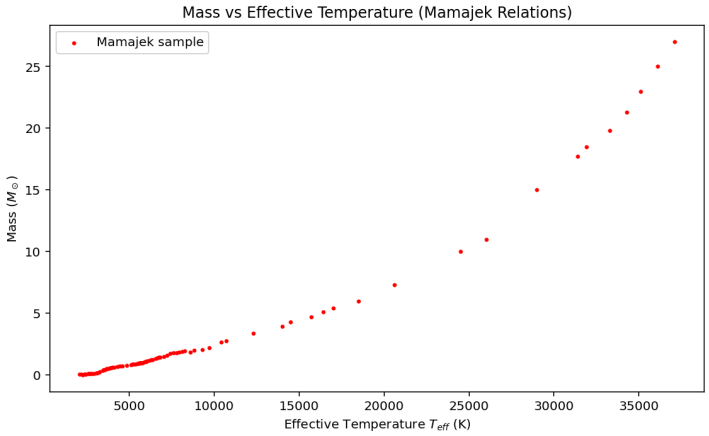


Figure 10: Stellar mass versus effective temperature $T_{\rm eff}$ from the Mamajek dataset.

93,407 have a $T_{\rm eff}$ from Gaia, whereas 73,613 dont. A mass estimate was obtained from another interpolation on the Mamajek $B_P - R_P$ colour-mass relation for these stars. This gave a mass estimate for 72,098 sources. These were combined which left 1,515 sources with no mass estimate. The relative impulse delivered to Oort Cloud comets by each stellar flyby was then computed. The majority of which was close to 0, however there were 2 values of ≈ 1.5 and ≈ 17.
Figures 11 and 12 show the perihelion time vs distance diagram for stars within 2 pc. The symbol size and colour are proportional to the impulse.

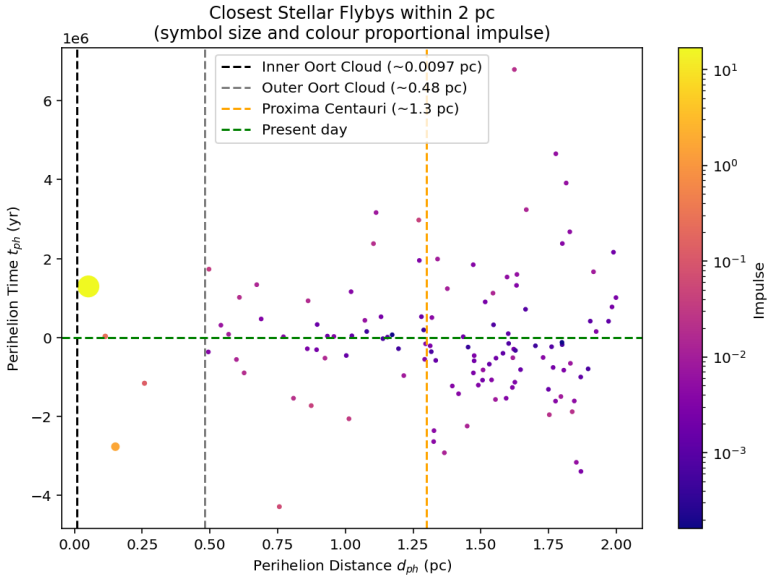


Figure 11: Perihelion time versus distance within $d_{ph} < 2\,\mathrm{pc}$, without error bars. Symbol size and colour are both proportional to the relative impulse. The large yellow spot has highest impulse by far (≈ 17).

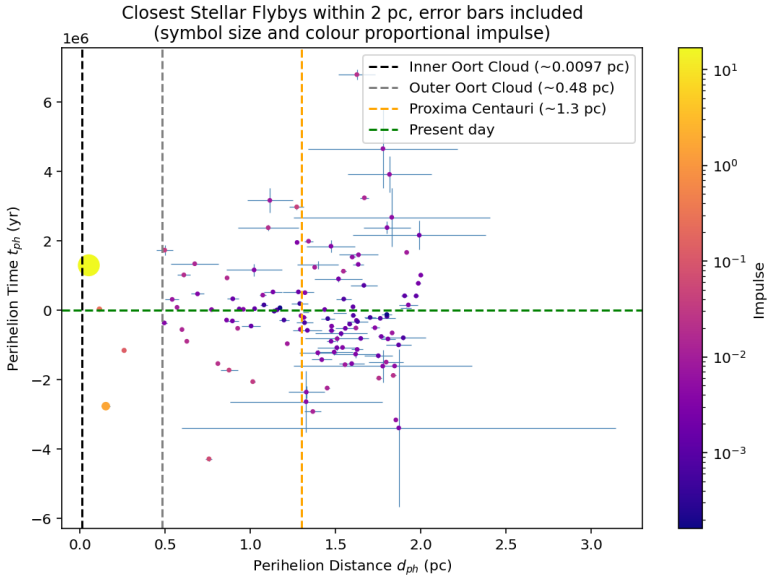


Figure 12: Same as Figure 11 but with error bars included.

The impulse plot highlights how a stellar flybys impulse is mostly affected by its perihelion distance, aligning with $I \propto d_{ph}^{-2}$. The star with the highest impulse is also the closest approaching star. So it is Gaia DR3 source 4270814637616488064, at 19.1 pc and with a perihelion distance of 0.051 pc. So its well within the outer Oort Cloud. Its d_{ph}^{-2} factor is much bigger than for stars passing at distances of $\approx$1 pc. This shows that a rare, very close flyby can have a huge impact on the Oort Cloud. Possibly bigger than the effect of many distant flybys.

Part 3

The Ultracool Sheet (Best et al. 2020) contains 3,890 objects. Most of these arent in the GCNS because their effective temperatures are below 2500 K which give almost no flux in the optical wavelengths which is where Gaia operates.
Figure 13 is a colour-magnitude diagram of the Ultracool Sheet. It plots the absolute J-band magnitude M_J against $J-K$ colour. The four spectral classes are colour coded: M dwarfs (red, $6 \leq$ sptnum < 10), L dwarfs (dark red, $10 \leq$ sptnum < 20), T dwarfs (sky blue, $20 \leq$ sptnum < 30), Y dwarfs (purple, sptnum ≥ 30). Directly imaged companions and exoplanets (gold) are scattered across the diagram.

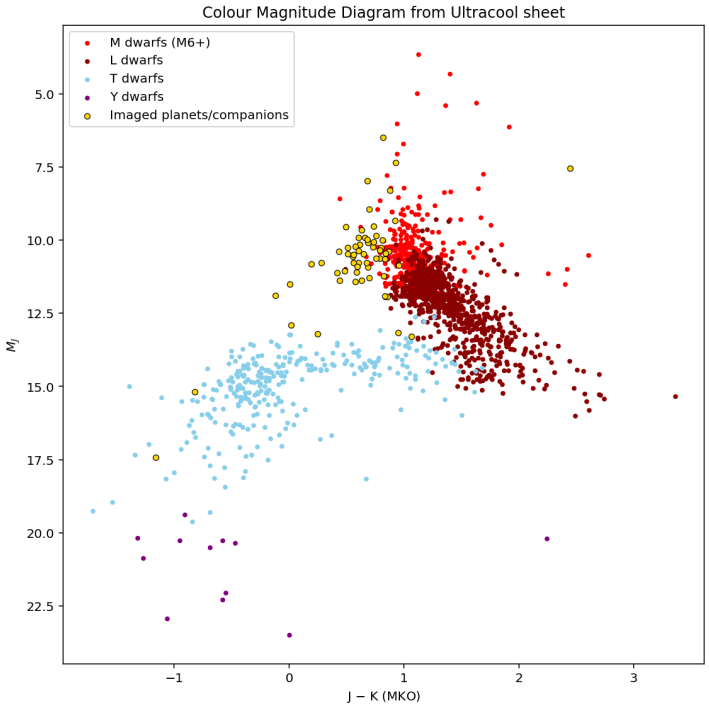


Figure 13: Colour-magnitude diagram for the Ultracool Sheet. M_J vs $J-K$ colour. Star types are colour coded by spectral class.

871 of the objects in the Ultracool sheet have measured radial velocities. This shows the difficulty of getting near infrared spectra of faint objects. This is why many of these objects arent in the GCNS, each spectrum has to be found individually with ground based observations.

Figures 14 and 15 show the t_{ph} versus d_{ph} plot for the Ultracool Sheet zoomed in for distances within 2pc. There is only a small number of close approachers compared to the GCNS sample. Monte Carlo uncertainties were done the same way as in Part 1. It seems that 2 sources are missing error bars, however they may be to small to see.

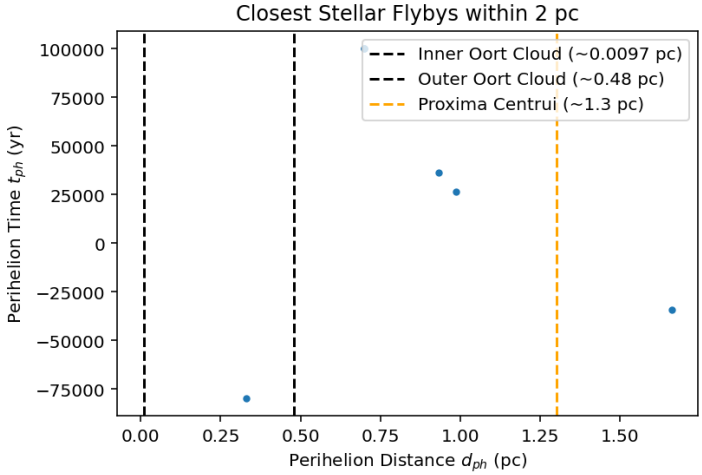


Figure 14: Perihelion time versus distance for the Ultracool Sheet sources within $d_{ph} < 2\,\mathrm{pc}$ without error bars. The same reference lines as before are still plotted

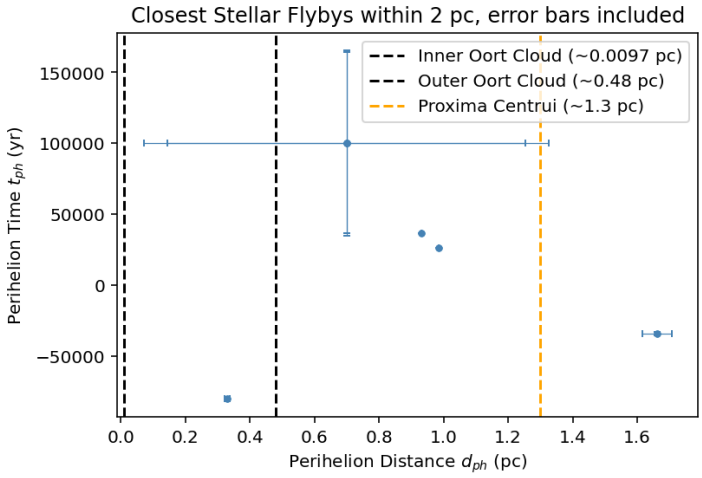


Figure 15: Same as Figure 14 with error bars included(¡100%).

The closest approaching ultracool dwarf is WISE J072003.20-084651.2, with $d_{ph} = 0.3298\,\mathrm{pc}$ and $t_{ph} = -79{,}801\,\mathrm{yr}$, so its closest approach happened around 80000 years ago. This is Scholz's star, it is well inside the Oort cloud. It possibly affected many comets within Oort Cloud in the past. Its spectral type number sptnum = 9.5 which matches with a spectral type of M dwarf but is very close to an L dwarf.
Figure 16 shows the closest approaching source overplotted on the colour mag-

nitude diagram. Its position within the M and L dwarfs is consistent an M9.5 spectral number at the edge of the L dwarf sequence. Just looking at the position, an L dwarf is the most likely, but the catalogued spectral number of 9.5 confirms its technically an M dwarf.

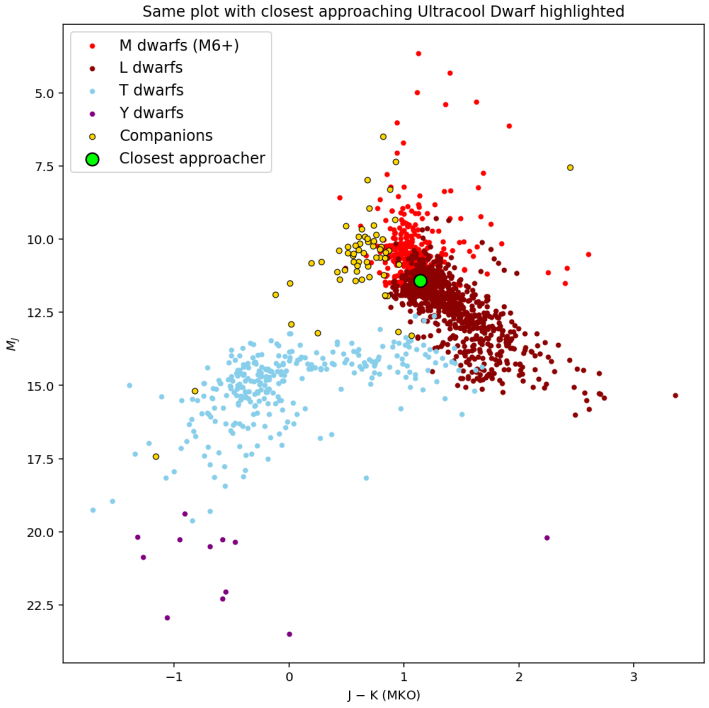


Figure 16: The same colour magnitude diagram as in Figure 13 but with the closest approaching ultracool dwarf (WISE J072003.20−084651.2) highlighted with the green dot.

5 Conclusions

Part 1:
The GCNS '100pc_rvnotnull.fits' file has 167020 sources with radialvelocity measurements. Monte Carlo error propagation with $N = 1000$ iterations was used to find perihelion times and distances and their uncertainties of every source. Gaussian input and output histograms for three sources (indices 0, 300, and 1000) verified the linear motion approximations. Within a ± 5 Myr window, 44 stars are within Proxima Centauri (1.3 pc), 4 within the outer Oort Cloud (< 0.48 pc) and none are closer than the inner Oort Cloud (< 0.0097 pc). The closest encounter is from source 4270814637616488064, it has a perihelion distance of $d_{ph} = 0.051 \pm 0.003$ pc and will happen in $t_{ph} = +1.295 \times 10^6 \pm 2.24 \times 10^4$ yr.

The ten closest approaching sources are mainly K and M dwarf stars with one possible white dwarf (BP − RP ≈ 0.8, $M_G \approx 14$).

Part 2:
Stellar masses were estimated using linear interpolation of the Mamajek text file (Pecault & Mamajek et al. 2013). There were 167020 sources and 93407 had a T_{eff}, which were used to get a temperature based mass estimate. For the other 73613 sources, a colour interpolation was used to get masses for another 72098 stars to cover the ones that weren't covered from the temperature based mass estimate. This left 1515 sources with no mass estimate.

The relative impulse imparted on the Oort Cloud was calculated. The vast majority of the total impulse from all stars in the catalogue came from the closest approaching star. This was source: 4270814637616488064, with a relative impulse of around 17. The second highest value was roughly 1.5 with the rest all far below 1. This aligns with the fact that the d_{ph}^{-2} term dominates the equation, therefore the distance is the biggest factor in a stars impulse imparted on the comets.

Part 3:
The UltraCool Sheet contains 3890 objects of spectral types M, L, T and Y. 871 of them have a measured radial velocity. This is because Gaia struggles to pick up the faint, cool objects at near infrared wavelengths. The closest approaching ultracool dwarf is Scholz's Star (WISE J072003.20-084651.2). It has $d_{ph} = 0.3298\,\text{pc}$ and $t_{ph} = -79{,}801\,\text{yr}$, so its closest approach happened around 80000 years ago. At its closest point, it was well within the Oort Cloud which could have caused a significant disturbance in its comets. It is a spectral type of 9.5, this puts it at the boundary between M and L dwarfs, however on the colour magnitude diagram it is moreso within the L dwarfs.

For some sources there was a calculated error of much higher than 100% which made the plots unreadable whne the error bars were applied. To fix this any error of more than 100% was not displayed on the graphs.

Overall, the analysis identified at least one flyby that will happen in the future (4270814637616488064) that will travel within the outer Oort Cloud, this will happen in $\approx 1.3\,\text{Myr}$. This is close enough to cause a comet shower. One flyby in the past of an ultracool dwarf (Scholz's Star) that could have caused a comet shower from the Oort Cloud approximately 80000 years ago.

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